

CELLCHECK · DESIGN DECISIONS REGISTER

CellCheck design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CCK-DEC-001	Version	0.2
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note and decision records, the build plan work and every decision made so far

CellCheck design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Decisions still to be made.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	How the guard is held down (part of the containment arrangement)	(a) two knurled thumb screws, as modelled; (b) two toggle latches; (c) a hinge at the back with one latch	(a) for the first prototype; decide for later units after the TRL 4 containment test	Guard flanges, rivet nuts (build plan sections 3.8 and step 11)	CCK-DDR-003, A1
2	Whether the wire slots under the guard are sealed after wiring	(a) open with grommet strip, as modelled; (b) filled round the leads with high-temperature silicone	(b)	Wiring step 9	CCK-DDR-003, A2
3	First trial partner	Repair café, e-bike repair shop or battery collection point	None stated	Not part of the build; sets the cells used at TRL 4	CCK-DDR-001, item 10
4	Scheduling rule for the attended day	(a) start a current stage only if it finishes the same day (12.0 cells per day); (b) pause overnight with the unit off (12.8 cells per day)	None made; (b) adds an unquantified capacity error	Firmware sketch only	CCK-DDR-001, item 12
5	Responses to the requirements at risk, R11 (mass, 4.91 kg against 5 kg) and R5 (throughput)	For R11: accept and weigh at TRL 4, or a 1.2 mm tray (saves about 0.49 kg but thins the containment); for R5: scheduling rule, more channels later	None made	Tray thickness if R11 changes	CCK-DDR-001, item 14; CCK-CAL-001 section 8
6	Grade A resistance limit per cell model (60 mΩ default)	Values per cell model from named data sheets	None; needs data sheets	Grading table only	CCK-DDR-001, item 16
7	18650 cells shown in the product renders where the model uses the 21700 envelope	Accept for the renders, or render 21700 cells	Accept; the model keeps the 21700 envelope for clearance checks	None	REVIEW 2026-09-26
8	Bins for matched groups A, B and C shown in the renders	(a) render context only; (b) a BOM line for three bins, costed against the value-engineering target	(a) now; decide (b) with TRL 4 purchasing	None now	REVIEW 2026-09-26
9	Fan finger guards shown in the renders	Add to BOM line 10 at TRL 4 and cost them then, or leave out	Add at TRL 4	Fans (step 7)	REVIEW 2026-09-26
10	Guard perforation drawn as slots and 1.0 mm sheet in the renders	Accept as appearance, or redraw with round holes and 0.8 mm	Accept; the BOM specification stays as written	None	REVIEW 2026-09-26

#	Decision needed	Options	Recommendation	Affects in the build	Source
11	Second-life SwapCell pack for PowerBox (low-current storage, about 334 Wh, 165 W)	Agree with the SwapCell and PowerBox projects, or not	Raise with those projects (cross-repo action)	None	CCK-DDR-001, item 1; CCK-CAL-001 section 10

■ To confirm when parts are bought

Table 2. Items to check against the parts before building.

#	What to confirm	Why it matters	Source
1	The steel tray is plain steel, about 460 x 300 x 45 mm with 1.5 mm walls, and its flat floor is at least 457 x 297 mm inside	The liner, plate and containment estimate assume it	CCK-DDR-003, P1
2	The rubber feet have M4 studs about 8 mm long	The stud must pass the 1.5 mm floor and engage the standoff	CCK-DDR-003, P1
3	The M4 rivet nuts grip 1.5 mm sheet and their heads are under 1 mm thick	The guard flange sits round the head	CCK-DDR-003, P4
4	The cell holders have two M3 fixing holes in their base and take 18650 and 21700 cells up to 70 mm	Holder screws and the 84 mm holder length	CCK-DDR-003, P3
5	The charger modules, current monitor breakouts, controller and converter have M3 mounting holes (or are fixed to a small perfboard that has them)	Each board stands on two nylon standoffs	CCK-DDR-003, P5
6	The 2.8 in display module is about 86 x 50 mm with four corner holes	Display stand size	CCK-DDR-003, P6
7	The heatsink spine is 6 mm thick, 50 mm tall, and has a fin gap at 20 mm from the left end and every 40 mm after it	MOSFET screws go between fins; plate and shroud screws are tapped into the spine	CCK-DDR-003, P7
8	The fans have a 50 mm square screw pattern	Shroud holes	CCK-DDR-003, P8
9	The DC jack, fuse holder and rocker switch cut-out sizes against the inlet housing (11.2 mm, 12 mm, 12 x 16 mm)	Inlet housing holes	CCK-DDR-003, P10
10	A sourced figure for the heat released by one 21700 cell in thermal runaway, to replace the assumed 100 kJ	Containment estimate before any TRL 4 work	CCK-CAL-001 section 7

■ Value engineering

Value-engineering target: USD 175.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 172.40 (USD 2.60 under the target).

Main cost drivers (CCK-CAL-001, section 11): the per-channel parts, which are bought eight times (cell holder with four-wire contacts, charger module, channel board, load MOSFET and op-amp loop, thermistor: about \$66 for the eight channels), the mains adapter (\$15), the steel tray with liner (\$12) and the controller board (\$10). The parts added for construction (lines 19 to 21) cost \$8.40.

Savings worth trying, should indicative prices rise above the target:

- Leave the matched-group bins and the fan finger guards out of the BOM until TRL 4 purchasing (open decisions 8 and 9).
- Look for a cheaper cell holder or charger module, since each saving is multiplied by eight.

■ Decisions made

Table 3. Decisions made, with dates and records.

Date	Decision	Decided by	Record
2026-09-25	Keep SwapCell in the pitch and plan a second-life variant; charge and discharge attended only (rest stage may be left); hardware watchdog and undervoltage comparators; grading thresholds, 14-day rest and ± 1 % group tolerance as starting values; TP5100, INA226 and ESP32-S3 class modules; eight channels with a linear load; LFP profile later; per-cell CSV mapped to the ReflowEconomy passport (items 1 to 3 and 5 to 9)	Amish: "i accept all your recommendations, go with them across all repos."	CCK-DDR-001, CCK-DDR-002
2026-09-25	Budget raised from \$160 to \$175 (item 4)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-09-25	DC four-wire resistance and a 14-day rest (item 11); firmware and build rules: same-channel cohort self-discharge reading, temperature-gated resistance pulse, per-channel calibration, fan-fault stop, detached-thermistor check (item 13); raise the passport gaps with ReflowEconomy (item 15)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-10-01	Design for construction: feet and standoffs, 1.5 mm plate on six standoffs, holder fixings and wire slots, folded guard with thumb screws, board standoffs, larger display stand, heatsink and shroud fixings, thermistor C-clips, inlet housing, 5 V converter and channel boards (P1 to P12)	Made under Amish's 2026-09-30 instruction: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." Open for his review	CCK-DDR-003